

Beyond the Syllabus

Approximation and Local Search: How Close Is Close Enough?

Week 14 Supplement – TOAE209/TOAH209: Design and Analysis of Algorithms

Foreign Trade University

Outline

Why look beyond the syllabus?

- This week's unifying lesson was the **bounding technique**: never compute OPT, but squeeze the algorithm's value against a quantity you *can* reason about (a matching, an MST, the triangle inequality).
- The lecture notes themselves flag a loose end: the MST-doubling tour is a 2-approximation for Metric TSP, and a remark says Christofides refines this to $\frac{3}{2}$ – “we do not pursue it here.” Three questions a curious student should ask next:
 - ① Christofides' $\frac{3}{2}$ stood unbeaten for over 40 years. Did anyone ever beat it?
 - ② This week's Vertex Cover 2-approximation is dead simple. Is 2 actually the best *any* efficient algorithm could ever do?
 - ③ Max-Cut local search is a 2-approximation that always terminates – but *how fast*, and why does local search work so well in practice on problems with no such guarantee at all?

How this supplement is different from a survey

Every section is sourced to a specific, verified paper (venue, year, and an arXiv/DOI identifier), the same standard used in the Week 2 and Week 3 supplements.

Recap: the ratios you have, in order

Algorithm	Ratio	Source
MST doubling + shortcut	2	this week's lecture
Christofides–Serdyukov (odd-vertex matching)	$\frac{3}{2}$	1976
?	?	open for ~ 44 years

For over four decades, no one could push the ratio below $\frac{3}{2}$ for *general* metric TSP, even though nobody could prove $\frac{3}{2}$ was optimal either (the best unconditional hardness result only rules out ratios below about 1.006).

The 2021 breakthrough

Karlin, Klein & Oveis Gharan, *STOC 2021* (Best Paper Award)

For some $\varepsilon > 10^{-36}$, there is a randomized $(\frac{3}{2} - \varepsilon)$ -approximation algorithm for metric TSP – the first improvement over Christofides in 44 years. arXiv:2007.01409; journal version in *Operations Research* 72(6):2543–2594, 2024.

- **Idea, at a very high level:** instead of picking *one* spanning tree (as the MST-based algorithm does) or one matching (as Christofides does), sample a spanning tree from a carefully chosen **max-entropy probability distribution** over spanning trees, then patch it into a tour the Christofides way.
- The ε is astronomically small ($> 10^{-36}$) – this is a *feasibility* result (“the ratio $\frac{3}{2}$ is *not* a hard wall”) far more than a practical algorithm. It reopened a problem many researchers had started to suspect was permanently stuck.
- A 2022 follow-up (arXiv:2212.06296) derandomized the result, giving a *deterministic* $(\frac{3}{2} - \varepsilon)$ -approximation.

Why this connects to next week

- Notice what did the work: **randomness**. Sampling a spanning tree from a max-entropy distribution – rather than deterministically taking the MST – is what breaks through $\frac{3}{2}$.
- This is exactly the pattern next week's lecture previews for Max-Cut: “a coin flip alone gives a $\frac{1}{2}$ -approximation *in expectation*, complementing this week's deterministic local-search guarantee.” Randomization is not a minor trick here – in both cases it is what an exhaustive 40-year deterministic search could not find.

Two kinds of “can we do better than 2?”

- This week's Vertex Cover algorithm (take both endpoints of every edge of a maximal matching) is a clean 2-approximation. Two very different questions ask “is 2 optimal?”:
 - ① **Unconditional hardness:** can we prove, from $P \neq NP$ alone, that no efficient algorithm beats some ratio? Dinur & Safra (2002) proved Vertex Cover cannot be approximated below 1.3606 unless $P = NP$ – real, but far from ruling out ratios between 1.36 and 2.
 - ② **Conditional hardness under a second conjecture:** assume something *stronger* than $P \neq NP$, and see how much further the wall moves.

The Unique Games Conjecture (UGC)

Khot, *STOC 2002*

The **Unique Games Conjecture** posits that a certain constraint-satisfaction problem (label each vertex of a graph so that, for each edge, a specified bijection between endpoint-labels is respected) is NP-hard to approximate even *extremely* well. It remains **open** as of 2026 – one of the most consequential unresolved questions in theoretical computer science, on the same tier as P vs. NP.

- If true, UGC implies a large collection of matching upper/lower approximation bounds across many NP-hard problems – turning “the best known algorithm” into “provably the best *possible* algorithm” for each of them.

Vertex Cover's 2 is (conditionally) exactly tight

Khot & Regev, *CCC 2003* / *J. Comput. Syst. Sci.* 2008

Assuming the Unique Games Conjecture, Vertex Cover cannot be approximated within $2 - \varepsilon$ for *any* fixed $\varepsilon > 0$, in polynomial time.

- Combine with this week's theorem: the maximal-matching algorithm achieves ratio 2, and (conditionally) **nothing can do better**. This week's simplest algorithm in the whole approximation toolkit is, plausibly, already optimal.
- Contrast with Set Cover: this week's greedy $O(\log n)$ -approximation is *also* known to be optimal, but *unconditionally* (Feige, 1998, via a different technique) – Vertex Cover's optimality is the more fragile, UGC-dependent kind.

Recap: what this week proved about Max-Cut local search

- Every single-vertex-flip local optimum cuts $\geq m/2$ edges – a 2-approximation – and each improving flip increases an integer cut value bounded by m , so termination takes *at most* m flips.
- That worst-case bound of m flips is real, but for some other local-search problems the analogous bound is **exponential**: there exist instances (for problems in the complexity class **PLS**, “Polynomial Local Search”) where every improving-move sequence to a local optimum has exponential length, even though checking a single move is fast.
- So why does local search – including Max-Cut’s own natural neighbourhood – so consistently converge fast *in practice*, on real, non-adversarial inputs?

Smoothed analysis answers “fast in practice, slow in theory”

Angel, Bubeck, Peres & Wei / follow-up line; unifying result: *ICALP 2024 / Math. Oper. Res.* 2024 (arXiv:2211.07547)

“On the Smoothed Complexity of Combinatorial Local Search” gives a unifying framework: perturb the input instance slightly at random (e.g. edge weights get tiny random noise), then bound the *expected* number of improving moves. For the classic 2-FLIP local search on Max-Cut, this smoothed number of steps is **quasi-polynomial** – far below the exponential worst case.

- **Reading it correctly:** this does not contradict the existence of genuinely exponential worst-case instances – it says such instances are *fragile*: an infinitesimal, realistic perturbation almost always destroys them.
- This is exactly why this week’s lecture recommends **random restarts** and **simulated annealing** as practical remedies – randomizing the input or the search path is precisely the kind of perturbation smoothed analysis models.

Summary

- 1 **TSP:** Christofides' $\frac{3}{2}$ stood for 44 years until Karlin, Klein & Oveis Gharan (STOC 2021 Best Paper) broke it with a randomized max-entropy spanning-tree sampler, later derandomized (2022) – randomness, not a smarter deterministic rule, was the key.
- 2 **Vertex Cover:** this week's 2-approximation is (conditionally, via Khot & Regev 2003/2008 and the still-open Unique Games Conjecture) already the best *any* efficient algorithm can do – a rare case where the simplest algorithm in the course may be provably unbeatable.
- 3 **Local search:** Max-Cut's $\geq m/2$ guarantee terminates in $\leq m$ steps, but general local search can take exponentially many steps in the worst case; smoothed analysis (2022–2024) explains why real instances almost never hit that worst case.

Looking ahead

Next week (*Randomized Algorithms*) makes randomness itself the main tool: randomized rounding, a randomized min-cut algorithm, and a one-coin-flip $\frac{1}{2}$ -approximation for Max-Cut *in expectation*. You have already seen a preview of why this matters – this week's biggest open-problem-breaking result was itself a randomized algorithm.